

Assignment 1

COMP1000 Introduction to Computer Programming

Semester 1 2023

Due date: 2nd April 2023 (Sun), 11:55 pm.

Assignment marks: 10 (contributes 10% of overall unit assessment).

1 Introduction

In assignment 1 you will be programming a basic game that features a playable character and some simple mechanics such as collision detection. In the demo video we show you an example solution that features a spider as the playable character. The demo video provides an essential visual clarification of what is required. Please pay close attention to the demo video as well as read this specification thoroughly. Please also refer to the provided marking rubric for the weighting of each feature.

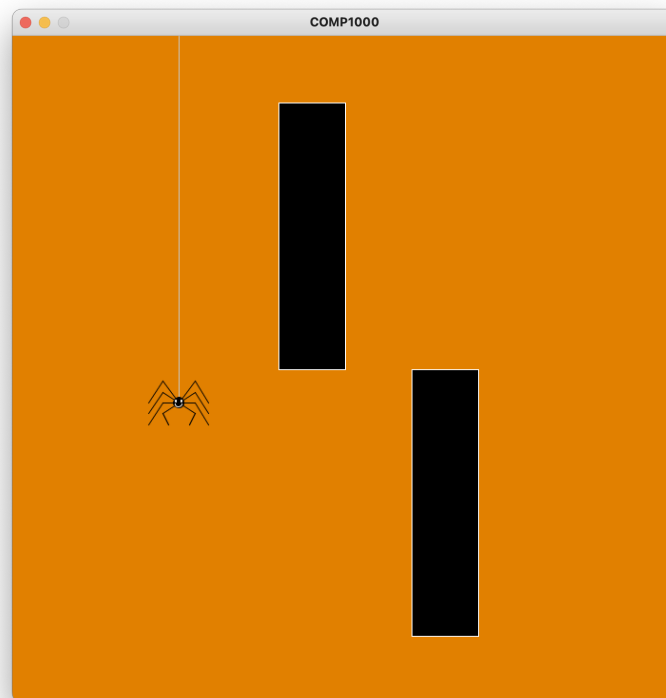


Figure 1: Example solution.

2 Assignment Specification

For full marks you need to implement all the following features correctly. You should submit just one program with as many of the features as you can get done.

Canvas

The canvas should be a square that is 700 pixels in width and 700 pixels in height.

Character design

You will need to come up with a design for your playable character. In the demo the playable character is a spider which is drawn using multiple lines and an ellipse. You are free to design your character however you like (or copy the spider design from the demo).

For full marks your character must be a complex shape drawn using at least 3 simple shapes. You must also use at least 2 different types of simple shapes. For example, a character that is drawn using 2 rectangles and a line OR a character drawn using two triangles and a circle would both suffice.

The character's overall dimensions should be approx. 70 pixels in width and 70 pixels in height.

Movement

Have the character be controlled using the arrow keys in the following manner:

- ☆ The UP arrow key moves the character up 70 pixels.
- ☆ DOWN moves the character down 70 pixels.
- ☆ LEFT moves the character to the left 70 pixels.
- ☆ RIGHT moves the character to the right 70 pixels.

The character should be centered on the point (35, 35) at the start of the program. This is to ensure that the character moves correctly. Please refer to the demo video to see what moving correctly looks like.

Note that you cannot use the WASD keys to move the character, the character MUST be controlled using the arrow keys.

Boundaries

The character must be prevented from moving off the canvas. For half marks, prevent the character from being moved beyond a canvas boundary.

For full marks, when the character is moved beyond a canvas boundary have the character appear against the opposite boundary. This is a recreation of the same boundary mechanic used in Pac-Man. Please see the demo video for a visual depiction of this feature.

Trail

There needs to be some kind of trail extending vertically from the character's position to the top of the canvas. In the demo there is a white line extending vertically from the spider to the top of the canvas to represent the web the spider is swinging from. You are free to change the theme of this trail, but it must behave in the same way. That is, it must extend vertically from the character to the top of the canvas AND it must move as the character moves.

Obstacles

Have two obstacles present somewhere on the canvas. Their positions should be randomly generated. This means that each time the program is run the obstacles will appear in different positions. You are free to design the obstacles however you like, but it's best to keep them rectangular.

There must be some visual indication that the game is over when the character collides with an obstacle. In the demo, the canvas is cleared and the text “game over” is displayed. You are free to design your own game over screen.

3 Constraints

- ✗ You cannot use any extra processing libraries or plugins.
- ✗ You cannot define your own classes.
- ✗ The only built-in java classes you can use are PVector and PShape. Although, we recommend not using them.
- ✗ You cannot use multiple tabs or multiple files.
- ✗ You cannot use transformations such as rotate, translate, scale etc.
- ✗ You cannot use images.
- ✗ You cannot import fonts.
- ✓ You must submit a single .pde file named exactly **COMP1000.pde** (not COMP 1000.pde, not COMP1000 (1).pde, not assignment1.pde...).

4 Marking Rubric

Character and canvas	None 0 marks	Character shape is present. Canvas size is 700 pixels by 700 pixels. 1 mark		
Movement	None 0 marks	Character moves in some fashion. 1 mark	Character is a complex shape and moves in some fashion. OR Character moves correctly with intended keypresses but is a simple shape. 2 marks	Character is a complex shape and moves correctly with intended keypresses. 3 marks
Boundaries	None 0 marks	Character does not move correctly with intended keypresses. 0 mark	Character cannot move off canvas. 1 mark	Moving beyond the canvas boundary brings the character to the opposite boundary. 2 marks
Trail	None 0 marks	There is a trail that extends vertically from the top border of the canvas to the character's position. This trail must move with the character. 1 mark		
Obstacles	None 0 marks	One obstacle is present. Its position is randomly generated. 0.5 marks	Two obstacles present, each with randomly generated positions. 1 mark	
Collisions	None 0 marks	One obstacle is present. When the character collides with the obstacle the game is over. 1 mark	Two obstacles are present. When the character collides with either obstacle the game is over. 2 marks	
Errors	Major error -4 marks	Error (includes incorrect filename) -2 marks	Small error -1 mark	No errors 0 marks